UNDER DUALIZING-COMPLEX, PRINCIPAL-IDEAL AND DEPTH-DIMENSION HYPOTHESES THE COMPLETION FUNCTOR ON FINITE LOCALLY FREE SHEAVES IS AN EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8089-8113 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 8115-8131, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al01 batch). Spans per computed label truth (al_clean_queue), not pool-card lines.

Editorial note. Cross-chapter homonym warning recorded in dedup (chapter-aware gate).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-equivalence-better' also exists in stacks-pione — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name inside the retained author proof become stub prereqs.

Source reference. Original lemma 0EJ9: sources/stacks-algebraization/pinned-original-source.tex, source lines 6736–6744. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EKU: sources/stacks-algebraization/pinned-original-source.tex, source lines 5062–5075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EJB: sources/stacks-algebraization/pinned-original-source.tex, source lines 6939–6957. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EJJ: sources/stacks-algebraization/pinned-original-source.tex, source lines 7480–7483. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EHC: sources/stacks-algebraization/pinned-original-source.tex, source lines 5200–5204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0EHC assume*

(1) *A has a dualizing complex and is I-adically complete,*
(2) *$I = (f)$ generated by a single element,*
(3) *A is local with maximal ideal $\mathfrak{a} = \mathfrak{m}$,*
(4) *one of the following is true*
   (a) *A_f is (S_2) and for $\mathfrak{p} \subset A$, $f \notin \mathfrak{p}$ minimal we have $\dim(A/\mathfrak{p}) \geq 4$, or*
   (b) *if $\mathfrak{p} \notin V(f)$ and $V(\mathfrak{p}) \cap V(f) \neq \{\mathfrak{m}\}$, then $\mathrm{depth}(A_\mathfrak{p}) + \dim(A/\mathfrak{p}) > 3$.*

Then with $U_0 = U \cap V(f)$ the completion functor

$$\mathrm{colim}_{U_0 \subset U' \subset U \text{ open}} Coh(\mathcal{O}_{U'}) \longrightarrow Coh(U, f\mathcal{O}_U)$$

is an equivalence on the full subcategories of finite locally free objects.

Proof. It follows from Lemma 0EKU that the functor is fully faithful (details omitted). Let us prove essential surjectivity. Let $(\mathcal{F}_n)$ be a finite locally free object of $Coh(U, f\mathcal{O}_U)$. By either Lemma 0EJ9 or Proposition 0EJJ there exists a coherent $\mathcal{O}_U$-module $\mathcal{F}$ such that $(\mathcal{F}_n)$ is the completion of $\mathcal{F}$. Namely, for the application of either result the only thing to check is that $(\mathcal{F}_n)$ satisfies the $(2, 3)$-inequalities. This is done in Lemma 0EJB. If $y \in U_0$, then the f-adic completion of the stalk $\mathcal{F}_y$ is isomorphic to a finite free module over the f-adic completion of $\mathcal{O}_{U,y}$. Hence $\mathcal{F}$ is finite locally free in an open neighbourhood U' of U_0. This finishes the proof. □